

John ‘Jack’ Sullivan

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EDUCATION

The Ohio State University

Ph.D. Aerospace Engineering

May 2026

Current GPA: 3.95/4.0

The Ohio State University

B.S. Aerospace Engineering. Magna Cum Laude, with Honors Research Distinction

August 2021

GPA: 3.76/4.0

COURSEWORK

Courses: Advanced Computational Fluid Dynamics, Fluid Mechanics, Numerical Methods for Hyperbolic Conservation Laws, Non-linear Systems Theory, Linear Systems Theory, Linear Mathematics in Finite and Infinite Dimensions, Hypersonics, Gas Dynamics, Heat Transfer, Numerical Methods for Linear and Non-Linear Systems

SKILLS

Languages: Fortran, Matlab, Bash, L^AT_EX

Tools: Direct Numerical Simulation, Large Eddy Simulation, Reynolds Averaged Navier-Stokes Simulations, MPI Programming, Data-Driven and Operator Based Modal Decompositions, Time-Frequency Analysis, Statistical Analysis, Linux

Numerical Methods Experience: High order finite difference schemes, characteristic boundary conditions, shock capturing approaches, finite volume schemes, finite element method, serial and parallel CPU codes

RESEARCH PROJECTS

Simulation of Spanwise Periodic Normal Shock Trains | *LES, F90, MPI, Matlab, Linux* May 2023 – Present

- Primary Ph.D. Thesis Project, using scale resolving simulations to analyze shock train unsteadiness.
- Augmented existing LES code by adding necessary boundary conditions, time marching, and damping procedures to simulate shock train flows.
- Created suites of analysis routines to quantify shock-boundary layer interaction dynamics for internal flow configurations, characterizing the evolution of near wall turbulence, wall pressure fluctuations, and streamwise shock oscillations occurring in the shock train.

Three-Dimensional Unsteadiness in a Shock Train Interaction | *LES, F90, MPI, Matlab* May 2024 – Present

- Secondary Ph.D. Thesis Project, using high-fidelity simulations to analyze three-dimensional effects of unsteadiness in a Mach 2.0 shock train in a low aspect ratio duct.
- Compare statistics of interaction to experiments at equivalent conditions to verify simulation accuracy.
- Determine how unsteadiness varies between shock/boundary layer interactions between the vertical (top and bottom) and lateral (side) walls.
- Determine why these changes manifest, using analyses of shock motions, boundary layer alterations, and wall pressure features.

Effect of Duct Aspect Ratio on Shock Train Interaction | *LES, F90, MPI, Matlab, Linux* Jan 2025 – Present

- Secondary Ph.D. Thesis Project, using scale resolving simulations to analyze effect of duct aspect ratio on structure of a Mach 2.0 shock train.
- Characterize the mean flow structure, with emphasis given to how separation and shock topology change as a function of aspect ratio.
- Also characterize the wall pressure signature, specifically focusing on changes to space time correlations and the scaling of observed frequency content

- Analysis of Moving STBLIs** | *LES, F90, MPI, Matlab, Linux* August 2020 – November 2023
- Tertiary Ph.D. Thesis Project, developed a data-driven technique to improve model order reduction and system identification for problems dominated by wave translation.
 - Method allows for frequency domain analysis of propagating shock wave/boundary layer interactions, with one possible application being analysis of scramjet unstart.
 - Completed as first graduate school project, this was also a continuation of work started during my bachelor's thesis.

- Self-Sustaining Unsteadiness in a Transonic Diffuser** | *DNS, F90, MPI, Matlab, Linux* May 2024 – Present
- Non-Thesis Research Project, using DNS level simulations to quantify dynamics of a shocked transonic diffuser.
 - Attention is given to the self-sustaining unsteadiness displayed by a normal shock/boundary layer interaction in a diffuser, by using a combination of non-linear simulations and linear operator-based decompositions.

- Parallelized mrDMD** | *F90, MPI, Linux* Spring 2022
- Developed a parallelized code capable of performing multi-resolution Dynamic Mode Decomposition.
 - Used the TSQR matrix algorithm for efficient singular value decomposition across an arbitrary number of processors.
 - Extracted tonal frequency content of a Mach 2 cavity flow at recursively finer time scales.

RESEARCH EXPERIENCE

- High-Fidelity Computational Multi-Physics Laboratory** | *Graduate Research Associate* Aug. 2021 – Present
- Graduate student position in Ohio State's Mechanical and Aerospace Engineering Department.
Conducting research using high fidelity CFD methods and advanced post-processing techniques.
Advised by Professor Datta V. Gaitonde.

- Air Force Research Laboratory** | *Computational Science Intern* Summers 2021 – Present
- Performed computational research for the Air Force, focusing on high speed flow configurations such as hypersonic cavities and scramjet isolators.

- Air Force Research Laboratory** | *Data Analysis Intern* Summer 2019
- Analyzed experimental data for turbine engine fluid-thermal systems.

PUBLICATIONS

Journal Articles

- Sullivan, Jack, and Gaitonde, Datta V., 'Turbulence Phenomena in a Shock Train Interaction', *Journal of Fluid Mechanics*, Accepted, Nov. 2025
- Sullivan, Jack, and Gaitonde, Datta V., 'A Modal Decomposition Method for Non-Stationary Phenomena: Application to Evolving Shock Trains', *Computers and Fluids*, Under Review.
- Sullivan, Jack, and Gaitonde, Datta V., 'Unsteady Mechanisms in a Shock Train Interaction', In Preparation.
- Sullivan, Jack, and Gaitonde, Datta V., 'Three-Dimensional Unsteadiness in Shock Train within a Low Aspect Ratio Duct', In Preparation.

Conference Papers

- Sullivan, Jack, and Gaitonde, Datta V., 'Effects of Duct Aspect Ratio on Shock Train Unsteadiness', AIAA Aviation 2026 Forum, In Preparation.
- Sullivan, Jack, and Gaitonde, Datta V., 'Dynamics of Shock Train in a Low Aspect Ratio Isolator' AIAA SciTech 2026 Forum, In Preparation
- Sullivan, Jack, and Datta V. Gaitonde. 'Unsteadiness in a Spanwise Homogeneous Shock Train Interaction' AIAA Aviation 2025 Forum. 2025.
- Sullivan, Jack, and Datta V. Gaitonde. 'Mean Flow and Turbulent Statistics of a Spanwise Homogeneous Shock Train.' AIAA SciTech 2025 Forum. 2025.

- Sullivan, Jack, and Gaitonde, Datta V., ‘Model Order Reduction of Scramjet Isolator Shock Dynamics During Unstart.’ ASME International Mechanical Engineering Congress and Exposition. Vol. 86656. American Society of Mechanical Engineers, 2022.
- Sullivan, John R., Erika Nosal, and Datta V. Gaitonde. ‘Application of Modal Decomposition to the Non-Stationary Scramjet Unstart Problem.’ ASCEND 2021. 2021.

Conference Proceedings, Invited Seminars, and Technical Briefings

- Sullivan, Jack, Hagenmaier, Mark, and Gaitonde, Datta, ‘Progress on Numerical Simulation of Shocked Transonic Diffusers’, Air Force Research Laboratory Outbrief (2025).
- Sullivan, Jack, and Gaitonde, Datta V., ‘Characterizing the Dynamics of a Spanwise Homogenous Shock Train using Large Eddy Simulation’, 50th Dayton-Cincinnati Aerospace Sciences Symposium (2025).
- Sullivan, Jack, and Gaitonde, Datta V., ‘The Interaction Between Shock Waves and Wall Turbulence in Hypersonic Engines’, Edward F. Hayes Advanced Research Forum (2025).
- Sullivan, Jack, Hagenmaier, Mark, and Gaitonde, Datta, ‘High-Fidelity Numerical Investigations into Three-Dimensional Isolators’, Air Force Research Laboratory Outbrief (2024).
- Sullivan, Jack, and Gaitonde, Datta V., ‘A Shift Procedure for Identifying Low Rank Behavior from Non-Stationary Dynamical System Data’, Bulletin of the American Physical Society (2023).
- Sullivan, Jack, Peterson, David, and Gaitonde, Datta, ‘Progress on Large Eddy Simulations of Spanwise Periodic Shock Trains’, Air Force Research Laboratory Outbrief (2023).

AWARDS

Conference and Competition Recognitions

- First Place, Art in Science Competition. 50th Dayton-Cincinnati Aerospace Sciences Symposium (2025).
- Second Place, Three Minute Thesis Competition. Ohio State Mechanical and Aerospace Engineering Graduate Showcase (2025).

ORGANIZATION INVOLVEMENT

Ohio State Mechanical and Aerospace Department Graduate Mentorship Program <i>Mentor</i>	2024-2025
Acted as mentor to a first year graduate student, helping them to get acquainted with Ohio State, graduate student life, and the city of Columbus, Ohio.	
Burgers Program Summer School on Turbulence <i>Attendee</i>	June 2024
Invited attendee to Burgers Summer Turbulence lecture series, hosted at University of Maryland	
American Institute of Aeronautics and Astronautics <i>Student Member</i>	2021-Present
Student member of the American Institute of Aeronautics and Astronautics (AIAA)	

MISCELLANEOUS

INTERESTS AND ACTIVITIES

Soccer, Squash, Chess, Biking, Hiking, Ancient and Classical History